Departmental LAN with VLANs

Cisco Packet Tracer Network Design & Configuration
Documentation

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Technology: Cisco Packet Tracer

Core Concepts: VLANs, 802.1Q Trunking, Router-on-a-Stick, Inter-VLAN Routing, IPv4

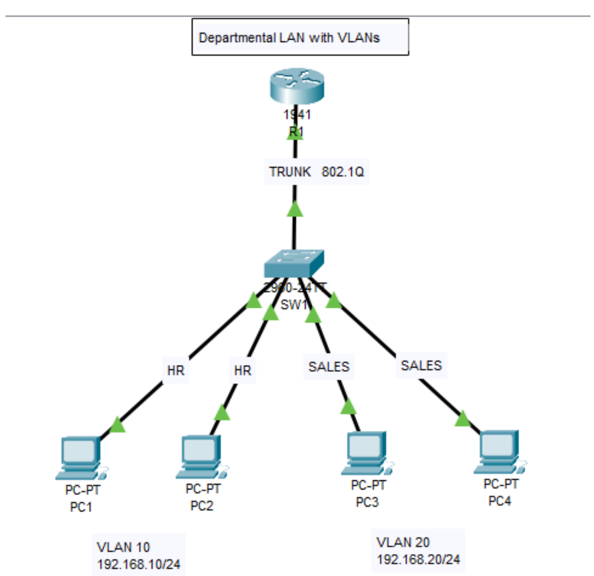
1. Project Overview

This project demonstrates the design and configuration of a small departmental enterprise LAN using Cisco Packet Tracer. The network separates HR and Sales users into different VLANs and IP subnets. A Cisco router provides Layer 3 communication between the VLANs using the Router-on-a-Stick architecture.

2. Project Objectives

• Design a departmental LAN topology.
• Create and configure VLAN 10 (HR) and VLAN 20 (Sales).
• Assign switch access ports to the correct VLANs.
• Configure an IEEE 802.1Q trunk between switch and router.
• Configure router subinterfaces for Inter-VLAN Routing.
• Configure default gateways and static IPv4 addresses.
• Verify Layer 2 and Layer 3 connectivity using Cisco IOS and ping tests.

3. Network Topology



4. Devices and Updated Port Connections

The physical wiring has been updated. The four PCs now use switch interfaces Fa0/2 through Fa0/5, respectively.

Device	Interface	Connection / Role
R1	G0/0	SW1 G0/1 – trunk
SW1	G0/1	R1 G0/0 – 802.1Q trunk
SW1	Fa0/2	PC1 – VLAN 10 (HR)
SW1	Fa0/3	PC2 – VLAN 10 (HR)
SW1	Fa0/4	PC3 – VLAN 20 (Sales)
SW1	Fa0/5	PC4 – VLAN 20 (Sales)

5. VLAN and IP Addressing Plan

VLAN	Department	Network	Gateway
10	HR	192.168.10.0/24	192.168.10.1
20	Sales	192.168.20.0/24	192.168.20.1

Device	VLAN	IP Address	Mask	Gateway
PC1	10	192.168.10.11	255.255.255.0	192.168.10.1
PC2	10	192.168.10.12	255.255.255.0	192.168.10.1
PC3	20	192.168.20.11	255.255.255.0	192.168.20.1
PC4	20	192.168.20.12	255.255.255.0	192.168.20.1

6. Switch Configuration

The Cisco 2960 switch is configured with two VLANs. Because the physical wiring was changed, the HR access ports are now Fa0/2–Fa0/3 and the Sales access ports are Fa0/4–Fa0/5. Gi0/1 remains the trunk toward the router.

```
enable
configure
terminal
hostname SW1

vlan 10 name HR exit

vlan 20 name SALES exit

interface
range fa0/2-3
switchport
mode access
switchport
access vlan
10 exit

interface
range fa0/4-5
switchport
mode access
switchport
access vlan
20 exit

interface
gigabitEthernet
0/1 switchport
mode trunk
exit

end
copy running-config startup-config
```

7. Updated Port-to-VLAN Mapping

VLAN 10 – HR: Fa0/2 → PC1, Fa0/3 → PC2

VLAN 20 – Sales: Fa0/4 → PC3, Fa0/5 → PC4

Trunk: Gi0/1 → R1 G0/0

8. Router-on-a-Stick Configuration

The router configuration is unchanged because the physical change was made only to the switch access-port wiring. The router continues to use G0/0.10 for VLAN 10 and G0/0.20 for VLAN 20.

```
enable
configure
terminal
hostname R1

interface
gigabitEthernet
0/0 no shutdown
exit

interface gigabitEthernet
0/0.10 encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
exit

interface gigabitEthernet
0/0.20 encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit


end
copy running-config startup-config
```

9. How Inter-VLAN Routing Works

A host in VLAN 10 sends traffic for VLAN 20 to its default gateway, 192.168.10.1. The switch forwards the VLAN-tagged traffic over the 802.1Q trunk to R1. The router routes the packet through G0/0.20 toward the 192.168.20.0/24 network. SW1 then forwards the frame through the appropriate VLAN 20 access port, now Fa0/4 or Fa0/5.

10. Verification and Connectivity Testing

Verify the updated access-port assignments:

```
show vlan brief

Expected:
VLAN 10 →
Fa0/2,
Fa0/3 VLAN
20 →Fa0/4,
Fa0/5
```

Recommended tests:
• PC1 (192.168.10.11) → PC2 (192.168.10.12): same VLAN connectivity.
• PC3 (192.168.20.11) → PC4 (192.168.20.12): same VLAN connectivity.
• PC1 → 192.168.10.1: HR gateway test.
• PC3 → 192.168.20.1: Sales gateway test.
• PC1 → 192.168.20.11: Inter-VLAN Routing test.
• PC3 → 192.168.10.11: reverse Inter-VLAN Routing test.

11. Verification Commands

```
SW1:
show vlan brief
show
interfaces
trunk show
interfaces
status show
mac
address-
table show
running-
config

R1:
show ip
interface
brief show ip
route
show running-config

PCs:
ipconfig
ping <destination-ip>
```

12. Troubleshooting Guide

Issue	Likely Cause	Check / Fix
Gateway unreachable	Wrong IP or gateway	Verify PC addressing
VLAN missing	VLAN not created	show vlan brief
Wrong VLAN	Access port misassigned	Confirm Fa0/2–Fa0/5 membership
Trunk failure	Port not trunking	show interfaces trunk

Subinterface down	Parent interface disabled	no shutdown on G0/0
Inter-VLAN ping fails	Wrong VLAN ID/gateway	Check dot1Q and gateway

13. Key Networking Concepts

VLAN: Creates a separate logical Layer 2 broadcast domain. **Access Port:** Carries traffic for a single VLAN toward an end device. **Trunk Port:** Carries traffic for multiple VLANs using VLAN tags.

IEEE 802.1Q: VLAN tagging method used on the trunk link.

Router-on-a-Stick: Uses router subinterfaces over a single physical interface to route between VLANs.

Default Gateway: Layer 3 address used by hosts to reach other networks.

14. Security and Design Note

VLANs provide logical segmentation and separate broadcast domains, but VLANs alone are not complete security controls. In a production network, ACLs, firewalls, port security, secure management protocols, monitoring, and appropriate access policies should be added.

15. Future Enhancements

- DHCP for automated addressing
- ACLs for department-level traffic control
- SSH for secure device management
- Switch port security
- Dedicated management VLAN
- Multiple switches and EtherChannel
- OSPF dynamic routing
- Layer 3 switching
- Wireless VLAN integration
- SNMP-based network monitoring

16. Learning Outcomes

This lab provides hands-on practice with Cisco IOS, VLAN creation, access-port assignment, 802.1Q trunking, IPv4 addressing, router subinterfaces, Inter-VLAN Routing, connectivity verification, and systematic network troubleshooting.

17. Project Deliverables

Recommended GitHub project structure:

- Departmental-LAN.pkt — Cisco Packet Tracer lab file
- README.md — project overview and configuration guide
- configurations/router-config.txt — router configuration

• configurations/switch-config.txt — updated switch configuration
• images/ — topology and verification screenshots
• documentation/Network-Design.pdf — technical documentation

18. Author

Tushar Patel

Aspiring Network & Cloud Engineer with hands-on experience in networking concepts, Cisco technologies, troubleshooting, and enterprise network design.